Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*System and Interface for Safe, Superhuman Robot Manipulator Teleoperation and Autonomy*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical · Materials		*Mechanical · Materials*	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/ Automation Machinery	*Robotics/Automation Machinery Software*	*Robotics/ Automation Machinery*	*Robot Control and Intelligence Technologies*
Concept of Proposed Technology	**Core Concept:** We aim to develop a safe, intuitive teleoperation system that allows a human operator to control a superhuman-strength robotic manipulator (e.g., 50–100 kg payload, tool speed > 10 m/s) with natural, low-latency motions. Teleoperation data will then be used to train safe autonomous manipulation policies, enabling robots to eventually perform difficult or dangerous tasks independently.			

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Why This Matters to Industry (Key Differentiator):

Most industrial robots today require fencing and fixed programming. Meanwhile, collaborative robots are safe, but weak and slow. There is no commercially viable product that combines:

- high power
- safety around humans
- intuitive control
- autonomous task execution

Background: We seek to build off of our recent work (see Fig. 1) on Shoulder-Elbow-Wrist (SEW) Mimic teleoperation, a new state-of-the-art for humanoid robot teleoperation with augmented reality (AR) [1], and our provably-safe manipulation and robot control [2-4].

Figure 1. PI Kousik's team has recently set a new state of the art for safe, immersive teleoperation of humanoid robots, shown here with a Rainbow Robotics RBY. See our project page: https://sew-geometric-teleop.pages.dev/

Rationale for Joint R&D Project

Summary: A joint R&D project is essential because the research institute contributes advanced algorithms, safety methods, and experimental prototyping, while the industrial partner contributes manufacturable hardware, reliability engineering, and commercialization pathways. Together, these capabilities enable the development of a first-to-market system for safe superhuman robot manipulation.

Benefits to Industrial Partner:

- **First access** to cutting-edge robot safety algorithms, intent prediction models, and teleoperation interfaces that competitors do not have.
- **De-risked development** through academic prototyping before expensive industrial-scale engineering occurs.
- **New product category creation:** safe, high-power general-purpose manipulators for construction, logistics, disaster response, and home-support markets.
- **Early data for commercialization:** user studies, surveys, and deployment tests help the SME assess product-market fit while shaping the R&D process.

Benefits to Research Institute:

- Access to high-power manipulators unavailable in academic labs.
- Ability to validate safety and autonomy algorithms under real industrial conditions.
- Close partner to translate research prototypes into real-world products.

Key Words

1. Superhuman robotic manipulation
2. Zero-latency augmented reality (AR) user interface for robot control
3. AI-driven user and robot coordination

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	4. Safe teleoperation 5. Safe AI for autonomous robot manipulation

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level 3	End of the Project	Level 7
R&D Objectives	**Overall Objective:** Develop a **zero-latency, safe human-robot teleoperation interface** for controlling large, powerful manipulators, and use teleoperation data to train **safe autonomous policies** that enable superhuman strength, speed, and precision in real-world tasks. **Subobjective 1: AI-Enhanced Teleoperation Interface** Design an intuitive interface that: • maps human motion to robot motion with <100 ms latency, • predicts operator intent to eliminate perceived lag, • provides haptic or force feedback, and • supports precise alignment (“object snap”) for heavy or fast tasks. **Industry relevance:** Enables safe, powerful, real-time remote work in hazardous or labor-limited environments. **Subobjective 2: Robust Safety Filter for Teleoperation and Autonomy** Develop safety systems that: • enforce **zero collisions** with static obstacles, • detect unmodeled failures (e.g., tool slip, object deformation), • communicate safety interventions to the operator. **Industry relevance:** Reduces liability, downtime, and risk, essential for scaling to real markets. **Subobjective 3: Autonomous Superhuman Manipulation Policies** Use teleoperated data to train autonomous policies that: • execute tasks at **superhuman speed** (4× over demonstration speed) • perform **superhuman-strength** lifting and carrying, • recover from disturbances without operator intervention. **Industry relevance:** Enables semi-autonomous and autonomous operation once deployed, increasing value and reducing reliance on skilled teleoperators.			

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R&D Target Specifications	**Objective-Linked Target Specifications**
	To ensure measurable progress toward commercialization, target specifications are defined for each R&D objective and aligned with industry-relevant performance, safety, and scalability requirements.
	
	Target 1: Teleoperation Performance and Usability
	The teleoperation system will achieve:
	• **End-to-end latency < 100 ms** between human input and robot execution
	• Human-to-robot tracking accuracy of
	  ○ **< 1 cm Cartesian position error**
	  ○ **< 10° end-effector orientation error**
	• Stable haptic or force feedback sufficient for manipulating heavy or high-moment objects as assessed by **> 75% task success rate**
	• Minimal operator training time, with novice users capable of completing benchmark tasks after **< 30 minutes of practice**
	Industrial relevance: Low latency and intuitive control are essential for operator safety, productivity, and adoption in real-world industrial environments.
	
	Target 2: Safety and Reliability Guarantees
	The system will enforce safety constraints during both teleoperation and autonomous execution, achieving:
	• **Zero collisions** with known static obstacles during execution, following our formal safety guarantees for collision avoidance under bounded uncertainty [2-4]
	• Failure detection rate of unmodeled task degradation (e.g., object slip, deformation, spillage) with **< 1 undetected failure per hour during operation**
	• Clear operator feedback when safety constraints intervene or override commands such that there is **< 1 user intervention per minute during operation** (tested by a shared teleoperation experiment)
	Industrial relevance: These specifications directly address regulatory approval, liability reduction, and customer trust, which are key barriers to deploying powerful robots near people.
	
	Target 3: Autonomous Superhuman Manipulation
	Autonomous policies trained from teleoperation data will demonstrate:
	• At least **4× speedup** over human demonstration trajectories
	• Superhuman strength capability, including **> 50 kg payload** for pick-and-place tasks
	• Superhuman motion execution, including **peak end-effector speed up to 10 m/s**
	• Superhuman precision, including **± 1 mm repeatability** during teleoperation
	• Recovery from disturbances without human intervention following above failure and intervention metrics
	Industrial relevance: These specifications enable high-value autonomous operation modes that differentiate the product beyond manual teleoperation alone.

R&D Plan	**R&D Plan Overview** The project follows a three-year staged development plan, progressively transitioning from academic prototyping to industry-ready hardware and software suitable for commercialization. Each year delivers standalone value to the industrial partner while building toward an integrated product. **Year 1: Teleoperation Interface and Foundational Data Collection** Technical Focus: • Design and prototype an ergonomic, intuitive teleoperation interface capable of providing force or haptic feedback • Develop AI-based operator intent prediction to reduce perceived latency • Implement initial human-to-robot motion mapping for large manipulators

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- Collect high-quality teleoperation datasets across representative manipulation tasks
- Validate performance in simulation and laboratory-scale hardware

Key Deliverables:

- Functional teleoperation interface prototype
- Demonstrated <100 ms control latency
- First generation operator motion datasets
- Preliminary performance benchmarking

Industrial Value:

- Early validation of user experience and usability
- Reduced risk before committing to manufacturable designs
- Immediate insight into human-robot interface design choices

Year 2: Safety System Integration and Autonomous Policy Training

Technical Focus:

- Develop a formally verified safety filter for collision avoidance
- Extend safety mechanisms to detect unmodeled failures using AI-based perception and monitoring
- Design user feedback mechanisms for safety intervention transparency
- Train imitation-learning policies using teleoperation data collected with safety constraints active
- Evaluate the effect of safety enforcement on task efficiency and user trust

Key Deliverables:

- Safety filter with formal guarantees
- Integrated teleoperation + safety system
- Preliminary autonomous manipulation policies
- Quantitative safety and failure-rate performance metrics

Industrial Value:

- Safety architecture suitable for certification and regulatory review
- Demonstrated reliability essential for product development
- Differentiation from competitors lacking provable safety guarantees

Year 3: System Integration, Deployment, and Superhuman Demonstration

Technical Focus:

- Develop autonomous failure recovery strategies integrated with the safety filter
- Fully integrate teleoperation interface, predictive control, safety system, and recovery behaviors
- Train and refine autonomous policies capable of superhuman speed, strength, and precision
- Deploy the complete system on full-scale, high-power manipulator hardware
- Evaluate real-world performance with representative users and tasks

Key Deliverables:

- Unified teleoperation and autonomous manipulation system
- Demonstration of superhuman performance metrics
- System validated in realistic environments
- Documentation supporting commercialization and pilot customer trials

Industrial Value:

- Near-product-ready system suitable for customer pilots
- Clear roadmap to commercialization and market entry
- Opportunity for first-mover advantage in a new robotics category

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will lead **core technology innovation**, including algorithm development, safety guarantees, and experimental validation, ensuring the project achieves its technical objectives while reducing technical risk for commercialization. **Key Responsibilities** • Design and prototype the **teleoperation user interface**, including ergonomic hardware concepts and human-to-robot motion mapping • Develop **AI-based intent prediction and motion anticipation algorithms** to enable zero-latency teleoperation • Develop **formally verified safety filters** for collision avoidance and constrained robot motion • Develop **failure detection and recovery algorithms** for both teleoperated and autonomous operation • Collect high-quality **teleoperation datasets** and train autonomous manipulation policies using imitation learning • Perform **laboratory-scale testing and benchmarking** to validate performance against R&D target specifications • Lead dissemination of research outcomes through publications and technical documentation **Collaboration Focus** The research institute will work closely with the industrial partner to: • adapt research prototypes to realistic hardware constraints, • define evaluation metrics relevant to commercial deployment, • and ensure algorithms are compatible with industrial software and hardware stacks. **Value to the Industrial Partner:** The research institute absorbs high-risk, high-reward technical development, delivering validated algorithms and prototypes that significantly reduce the SME's development cost and time-to-market.
Expected R&R of Korean SME Partner	The Korean SME partner will lead **productization, system integration, and commercialization**, transforming research prototypes into reliable, deployable systems suitable for real customers and markets. **Key Responsibilities** • Design and develop **manufacturable teleoperation interface hardware**, scaling research prototypes into robust, cost-effective products • Integrate algorithms with **industrial-grade hardware**, embedded compute, and real-time control systems • Conduct **large-scale and real-world testing** beyond laboratory environments, including representative use cases and users • Provide **feedback on usability, reliability, and maintainability**, guiding iterative refinement of research outputs • Lead **customer discovery activities**, including focus groups and market analysis, to assess commercialization potential • Evaluate pathways for **certification, regulatory compliance, and deployment** **Collaboration Focus** The SME partner will work with the research institute to: • define practical system constraints early in development, • prioritize features that align with customer needs, • and ensure the final system meets both technical and market requirements.

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	Value to the Research Institute: The SME partner provides access to realistic deployment scenarios, industrial hardware, and commercialization expertise, enabling validation of research at a scale and fidelity not possible in academia alone.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technical Outcomes** The joint project aims to deliver: • A **fully integrated teleoperation system** enabling safe, low-latency control of large, high-power robotic manipulators • **Formally verified safety algorithms** that prevent collisions and mitigate unmodeled failures during operation • **Autonomous manipulation policies** capable of superhuman strength, speed, and precision • A new class of **high-quality teleoperation datasets** for training robust robot autonomy • A **TRL-7 validated system** demonstrated on full-scale hardware in realistic environments These outcomes represent a significant advancement over current robotic systems, which typically trade off safety, power, or autonomy. **Industrial and Commercial Impact** For the Korean SME partner, the project is expected to: • Enable creation of a **new product category**: safe, superhuman robotic manipulators operable near people • Provide **first-mover advantage** through early access to unique safety and teleoperation technologies • Reduce commercialization risk through early validation of usability, reliability, and customer demand • Support future **market expansion** into construction, disaster response, logistics, manufacturing, and assistive robotics • Establish a strong foundation for **pilot deployments and customer trials** immediately following the project **Broader Societal and Economic Impact** The proposed technology has the potential to: • Improve worker safety by removing humans from hazardous or physically demanding tasks • Address labor shortages in construction, logistics, and care work • Enable rapid response in disaster and emergency scenarios • Enhance quality of life through assistive robotic technologies By combining advanced research with industrial product development, the project supports both **technological leadership** and **economic competitiveness** for the participating Korean SME.

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6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization